MASTER PRD

Rivals

Group Habit Tracker with Live Leaderboards & Proof Logging

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"Your friends know if you showed up. Your streak knows if you faked it. Rivals makes sure neither can lie."

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1. Introduction & Executive Summary

1.1 Product Name

Rivals — Group Habit Tracker with Live Leaderboards & Proof Logging

1.2 Vision & Mission

To make personal discipline a shared sport. Rivals turns the lonely, forgettable act of building a daily habit into a live, group-wide competition where everyone is accountable to everyone else — in real time, with proof you cannot fake.

1.3 Executive Summary

Rivals is a cross-platform mobile and web application (React Native for iOS and Android, with full-parity web access) built around one core insight: habits stick when other people are watching. A user signs up, creates or joins groups, and the admin of each group sets one or more daily habits that every member must complete. Completion is not self-reported — it requires a live camera photo taken on the spot, with a tamper-proof timestamp watermark automatically burned into the image. No gallery access. No uploads from a photo taken yesterday. The moment you mark a habit complete, you have to prove you are there, right now.

Each group maintains a live leaderboard. The group admin chooses the leaderboard mode: ranked by current streak, by total completions, or by score within a time-boxed challenge window. Members see each other's progress, proof photos, and standing in a group activity feed. The social pressure of the leaderboard — combined with the integrity guarantee of the proof system — creates a uniquely honest accountability loop.

Rivals is **completely free for now.** No subscription tiers, no paywalls, no paid features. Every user has full access to everything for now. (v1.0)

1.4 Core Value Proposition

- **Accountability with teeth:** Proof photos and timestamp watermarks eliminate faking. Your streak reflects reality.
- **Group-first design:** Every feature is built around the group, not the individual. Habits mean more when others share them.

- **Live competition:** A real-time leaderboard turns mundane daily tasks into an ongoing contest with friends.
- **Zero setup friction:** No wearables, no integrations, no manual data entry. Just open the app, take the photo, done.

2. Strategic Goals & Market Opportunity

2.1 User Goals

- Build consistent daily habits through social accountability, not willpower alone.
- Along with group tracking, have an entire private section for your own personal habits like a regular habit tracker.
- See real-time proof that your friends are actually doing the work — not just claiming to.
- Compete meaningfully with people you know, on habits you all agree matter.
- Track personal progress over time: streaks, completions, historical logs.
- Access the app from any device — phone (iOS or Android) or browser — without losing anything.

2.3 Non-Goals (v1.0 specific)

- Rivals is **not** a social media platform — no public profiles, follower graphs, or content feeds outside groups.
- Rivals will **not** support non-daily habit frequencies in v1.0 (e.g., 3x per week).
- Rivals will **not** allow members to suggest or vote on habits; the admin has sole authority.
- Rivals will **not** have any paid features, subscriptions, or premium tiers.

2.4 Market Opportunity & Competitive Landscape

The habit-tracking app market is saturated with solo tools (Habitica, Streaks, Finch, Done). None of them solve the core problem: individual commitment is fragile. Group commitment, with visible proof and a live leaderboard, is a fundamentally different product category. The closest competitors are Bereal (timestamped photos, no habits), Strava (social fitness, no general habits), and various group fitness challenges (no proof system, no leaderboard). Rivals occupies a genuinely uncrowded space.

Competitor	What They Do	Rivals' Differentiation
Habitica	Gamified solo habit tracker	Rivals is group-first, real-time, and requires proof of completion. No self-reporting.

Competitor	What They Do	Rivals' Differentiation
Bereal	Timestamped candid photo sharing	Rivals ties photos to habit completion in a structured accountability loop, not social media.

Strava	Social fitness tracking	Rivals covers any habit (not just fitness) and works without wearables or third-party integrations.
WhatsApp / manual check-ins	Friends manually sharing progress in chat	Rivals automates tracking, eliminates manual reporting, and enforces proof. No spreadsheets.
Corporate wellness apps	Employer-run challenges, often paid/closed	Rivals is free, personal, and friend-group-native rather than top-down.

3. Target Audience & User Personas

Rivals targets people who have tried to build habits alone and failed — not because the habits were wrong, but because there was nobody watching. The app works for any habit that can be proven with a live photo: gym, running, meditation, reading, morning routines, cold showers, healthy meals, and more.

Persona 1: The Friend Group Organizer (Primary)

- **Name:** Aryan, 24
- **Profile:** College student. Runs a tight 6-person friend group obsessed with fitness. Currently uses a WhatsApp poll every Sunday to check who hit the gym that week. Half the group lies.
- **Pain Points:** No accountability. Self-reporting is unreliable. The spreadsheet he built gets ignored after week 2. He has no way to prove who actually showed up.
- **Job-to-be-Done:** "When I set up a group challenge with my friends, I want the app to force everyone to prove they did the work — no more honour system."
- **What Success Looks Like:** Creates a Rivals group, sets a daily gym habit, adds his friends by their usernames.

By week 2, everyone is competing fiercely and nobody wants to drop off the leaderboard.

Persona 2: The Reluctant Habit Builder

- **Name:** Shreya, 27
- **Profile:** Working professional. Has tried Habitica, Notion habit trackers, and paper journals.

Consistently drops off after 2 weeks with no external pressure.

- **Pain Points:** Solo tracking feels meaningless. Notifications are easy to ignore. There is no real consequence for missing a day.
- **Job-to-be-Done:** "When I feel like skipping, I want to know my friends will see that I skipped — and that is enough to make me go."
- **What Success Looks Like:** Joins a Rivals group with two close friends. The leaderboard keeps her honest for 30+ days straight — her longest ever streak.

Persona 3: The Competitive High-Achiever

- **Name:** Rohan, 30
- **Profile:** Gym regular, obsessive about personal bests. Doesn't need motivation, but wants competition. Currently number 1 in his friend group by a wide margin.

- **Pain Points:** Nobody around him is as consistent. Manual tracking is boring. He wants a real leaderboard with real proof.
- **Job-to-be-Done:** "I want to see, in real time, whether anyone is close to catching me — and I want their proof photos to confirm they actually earned it."
- **What Success Looks Like:** Checks the leaderboard every morning. Gets notified when someone closes the gap. Uses the challenge mode to set 30-day sprints with stakes.

4. Product Features & Requirements (v1.0)

Features are organized into Epics and assigned a priority: **Must-Have** (launch blocker), **High** (within 4 weeks post-launch), **Medium** (post-launch iteration), **Low** (future consideration).

EPIC 1: Authentication & User Identity

Every user has a globally unique username. Authentication is the gateway to all groups and habits.

ID	Feature	Description & Requirements	Priority
AUTH-101	Email / Password Sign-Up	Users register with a unique username, email address, and password. Username must be globally unique across the platform (case-insensitive), 3–20 characters, alphanumeric + underscores only. Email verification required before access is granted.	**Must-Have**
AUTH-102	Google Sign-In (OAuth 2.0)	One-tap Google Sign-In on mobile and web. On first Google login, user must choose and confirm a unique username before accessing the app. Subsequent logins use the stored username.	**Must-Have**
AUTH-103	Unique Username Enforcement	Username uniqueness is enforced at the database level with a unique constraint. Real-time availability check shown in the UI as the user types (debounced). Usernames cannot be changed after account creation in v1.0.	**Must-Have**
AUTH-104	Session Management	JWT-based sessions. Refresh tokens stored securely (iOS Keychain / Android Keystore / HttpOnly cookie for web). Auto-refresh before expiry. Log out from all devices option in account settings.	**Must-Have**
AUTH-105	Account Management	Change display name, change password, update avatar, delete account (with full data purge within 30 days), export personal data. Username and email changes deferred to post-v1.0.	**Must-Have**
AUTH-106	Password Reset	Email-based password reset flow with a time-limited secure token. Link expires after 15 minutes or first use.	**Must-Have**

EPIC 2: Groups & Membership

Groups are the fundamental unit of Rivals. A group is a fully isolated environment: its habits, logs, leaderboard, and activity feed are completely independent of any other group on the platform.

ID	Feature	Description & Requirements	Priority
GRP-201	Create Group	Any authenticated user can create a group. Required fields: group name (3–50 characters), optional group description, optional group avatar/cover image. Creator automatically becomes the group admin.	**Must-Have**
GRP-202	Invite by Username	Admin can search for any registered username and send them a group invite. The invited user receives an in-app notification and can accept or decline. Only the admin can invite members.	**Must-Have**
GRP-203	Invite via Link / Code	Each group generates a unique shareable invite link and a short alphanumeric code (e.g., RIV-4X9K). Tapping the link opens the app (or web) to a join confirmation screen. Links can be disabled by the admin. Deferred deep linking: link works even if app is not yet installed — user is directed to the store, and after install lands on the join screen.	**Must-Have**
GRP-204	Group Membership Cap	Maximum 100 members per group (including the admin). Attempting to join a full group shows a clear error message. Admin can remove members to free up space.	**Must-Have**
GRP-205	Admin Controls	Admin can: remove any member, edit group name/description/avatar, manage habits, change leaderboard mode, start/end challenge windows, and transfer admin role. Admin cannot leave the group without first transferring the admin role to another member — the Leave button is disabled until a transfer is made.	**Must-Have**
GRP-206	Member Controls	Regular members can leave a group at any time. Their historical logs are retained for group records but their active participation ends. They can rejoin only if re-invited by the admin.	**Must-Have**

GRP-207	Multi-Group Membership	A user can belong to any number of groups simultaneously. Each group is completely isolated: habits, logs, streaks, and leaderboards do not cross over. Completing a habit in Group A does not count toward Group B, even if the habit name is identical.	**Must-Have**
GRP-208	Group Discovery	In v1.0, groups are private by default and discoverable only via invite link or direct username invite. No public group directory in v1.0.	**Must-Have**

EPIC 3: Habit Management

Habits are the core unit of work in a group. The admin defines them; every member is expected to complete them daily. Completion requires both a checkbox acknowledgment AND a live proof photo.

ID	Feature	Description & Requirements	Priority
HAB-301	Create Habit (Admin Only)	Admin can create one or more habits for the group. Required fields: habit name (e.g., 'Morning Run'), optional description, optional grace period (0, 1, or 2 days). All habits are daily frequency in v1.0. A group can have multiple active habits simultaneously.	**Must-Have**
HAB-302	Grace Period Setting	Per habit, the admin can optionally allow a grace period of 1 or 2 days. If a user misses the habit on Day D, they have until the end of Day D+grace_period to log it. If they log within the grace window, their streak is fully protected and the missed day is forgiven — no retroactive logging required. If they do not log within the grace window, the streak resets to 0.	**Must-Have**
HAB-303	Edit / Deactivate Habit (Admin Only)	Admin can edit a habit's name, description, or grace period at any time. Editing affects only future logs; historical logs are not altered. Admin can deactivate a habit, which hides it from the logging UI but preserves all historical data. Deactivated habits can be reactivated.	**Must-Have**
HAB-304	Habit Completion: Dual Requirement	To mark a habit complete for the day, the user must: (1) tap the completion checkbox AND (2) take and submit a live proof photo. Both steps are mandatory. Skipping either step leaves the habit as 'incomplete' for that day. This dual requirement cannot be bypassed or disabled by any user, including the admin.	**Must-Have**

HAB-305	Same-Day Log Editing	A user may delete their log submission and re-submit within the same calendar day (in their local timezone). Once the day ends (midnight local time), the log is locked and cannot be altered, deleted, or backdated.	**Must-Have**
HAB-306	Multiple Habits per Group	A group can have multiple active habits. Each habit is tracked, logged, and scored independently. A user must complete all active habits to be considered fully compliant for that day. The leaderboard can aggregate across all habits in a group (admin configures this).	**Must-Have**

EPIC 4: Proof Photo & Anti-Cheat System

The proof photo system is the integrity backbone of Rivals. It is designed to make cheating meaningfully difficult while keeping the honest user experience fast and frictionless.

ID	Feature	Description & Requirements	Priority
PHT-401	Live Camera Only	When a user initiates habit proof, the app opens the device's native camera directly. Access to the photo gallery is explicitly blocked. The user must take a new photo in that moment. This is enforced at the OS permission level — the app requests camera permission only, never photo library permission for this flow.	**Must-Have**
PHT-402	Timestamp Watermark	Immediately after the photo is captured (before upload), the app renders a tamper-proof watermark onto the image. Watermark content: date (DD/MM/YYYY), time (HH:MM:SS), username, and habit name. Watermark is positioned in the bottom-left corner, semi-transparent white text with a dark stroke, at a fixed font size that is legible but not obstructive. The watermark is burned into the image pixels — it cannot be removed by the user.	**Must-Have**
PHT-403	Server-Side Timestamp Verification	The server records a server-side timestamp at the moment the upload request is received. If the difference between the client-reported timestamp (from the photo metadata) and the server-received timestamp exceeds a configurable threshold (default: 5 minutes), the upload is rejected with an error. This guards against device clock manipulation.	**Must-Have**

PHT-404	Photo Storage	Proof photos are stored in cloud object storage. Each photo is stored with: user_id, group_id, habit_id, log_date, client_timestamp, server_timestamp, and storage URL. Photos are stored for the lifetime of the group. On account deletion, user's photos are deleted within 90 days.	**Must-Have**
PHT-405	Group-Wide Photo Visibility	All proof photos submitted by any group member are visible to all members of that group. Viewing is accessible from the Group Activity Feed (inline thumbnail) and from any member's log detail view. Photos are not visible outside the group.	**Must-Have**
ID	**Feature**	**Description & Requirements**	**Priority**
PHT-406	Web Camera Support	On the web platform, habit proof uses the browser's native MediaDevices camera API (getUserMedia). Gallery upload is blocked. The timestamp watermark is applied client-side using HTML Canvas before upload. Web camera must be HTTPS-only (enforced by browser security model).	**Must-Have**
PHT-407	Photo Compression	Before upload, photos are compressed client-side to a maximum of 1 MB (target: 400–600 KB) while maintaining legible watermark quality. This reduces storage costs and speeds up feed loading.	**High**

EPIC 5: Leaderboard Engine

The leaderboard is the competitive heartbeat of each group. The admin selects one active leaderboard mode at a time and can switch modes at any point. All three modes are described in detail in Section 8. This epic covers the display and management requirements.

ID	Feature	Description & Requirements	Priority
LB-501	Three Leaderboard Modes	Admin selects one active mode per group: (1) Streak Mode — ranked by current streak. (2) Total Completions Mode — ranked by all-time completions since group creation. (3) Challenge Window Mode — ranked by completions within a time-boxed window set by the admin. Mode can be changed by the admin at any time.	**Must-Have**

LB-502	Live Leaderboard Display	The leaderboard screen shows: rank position, member avatar, display name, username, score The leaderboard screen shows: rank position, member avatar, display name, username, score (contextual to active mode), delta since last refresh, and the current user's row always highlighted. Data refreshes every 60 seconds via polling. Pull-to-refresh available for immediate update.	**Must-Have**
LB-503	Tie-Breaking Logic	Ties are resolved in order: (1) higher current streak, (2) higher longest-ever streak, (3) earlier join date (longer member gets preference). Tie-breaking logic is visible in a tooltip on the leaderboard.	**Must-Have**
LB-504	Mode Switch by Admin	Admin can change the leaderboard mode from the Group Settings screen at any time. Switching mode does not delete historical data from other modes. A banner is shown to all members when the mode is changed.	**Must-Have**

ID	Feature	Description & Requirements	Priority
LB-505	Challenge Window Management	In Challenge Window Mode: admin sets a name, start date, and end date for the window. During the window, members see a countdown timer and their current window score. At window end, a winner is declared and a 'Winner' badge is awarded. Historical windows are archived and viewable. Admin can start a new window after the previous one ends.	**Must-Have**
LB-506	Minimum Members Gate	A leaderboard with fewer than 2 members with at least one log shows a 'Waiting for Rivals' placeholder. The competitive display does not go live until at least 2 members have submitted at least one proof log.	**Must-Have**
LB-507	Multi-Habit Aggregation	For groups with multiple habits, the leaderboard score aggregates completions across all active habits. Each habit completion counts as 1 unit. In Streak Mode, a streak day requires ALL active habits to be completed (unless admin configures partial credit — deferred to post-v1.0).	**Must-Have**

EPIC 6: Group Activity Feed

The activity feed is the social heartbeat within each group. It is not a public social media feed — it is visible only to group members and contains only habit-related events. No comments, no likes, no user-generated text posts.

ID	Feature	Description & Requirements	Priority
FEED-601	Proof Photo Feed	Each completed habit log appears in the feed as a card: member avatar, display name, habit name, timestamp, and the proof photo as an inline thumbnail. Tapping the thumbnail opens the full-size photo. Cards appear in reverse chronological order (most recent first).	**Must-Have**
FEED-602	Event Types	Feed shows: habit completions (with photo), streak milestones (7-day, 30-day, etc.), member joined group, leaderboard mode changed by admin, challenge window started / ended / winner declared, member left group.	**Must-Have**
FEED-603	Real-Time Updates	New feed entries appear within 60 seconds via polling. A 'New activity' banner appears at the top of the feed when new items arrive while the user is viewing it. Pull-to-refresh triggers an immediate fetch.	**Must-Have**
FEED-605	Feed Pagination	Feed loads 20 items at a time. Infinite scroll loads older items on demand. Feed is available offline (read-only) showing the last cached page.	**Must-Have**

EPIC 7: Notifications

ID	Feature	Description & Requirements	Priority
NOTIF-701	Daily Habit Reminder	Push notification sent to users who have not yet completed their habit(s) for the day. Default send time: 8 PM local time. User can customize the reminder time or disable it per group. Reminder is not sent if all habits are already completed.	**Must-Have**

ID	Feature	Description & Requirements	Priority

NOTIF-702	Streak At Risk	If a user has a streak ≥ 3 days and has not completed today's habits by their grace window deadline, send a 'Your streak is at risk' push notification 2 hours before the deadline.	**Must-Have**
NOTIF-703	Leaderboard Position Change	Notify user when they are overtaken on the leaderboard (e.g., 'Rohan just moved above you!'). User can toggle this notification on/off.	**High**
NOTIF-704	Member Proof Submitted	Notify group members when a member submits a proof photo. User can choose: notify on every submission, notify only for members they follow (future), or disable.	**High**
NOTIF-705	Group Invite Received	In-app and push notification when a user receives a group invite from an admin.	**Must-Have**
NOTIF-706	Challenge Window Events	Notify all group members when a challenge window starts, when 24 hours remain, and when the window ends with the winner announcement.	**Must-Have**
NOTIF-707	Streak Milestone	Notify user (and optionally broadcast to the group feed) when they hit a streak milestone: 7 days, 14 days, 30 days, 60 days, 100 days.	**High**
NOTIF-708	Notification Preferences	Users can granularly control which notification types they receive, per group or globally. Preferences accessible from Profile settings.	**Must-Have**

EPIC 8: Gamification & Streaks

ID	Feature	Description & Requirements	Priority
GAM-801	Streak Tracking	Each user has a per-habit, per-group streak: consecutive days on which the habit was completed (proof photo submitted and accepted). Streak increments at day rollover if the habit was completed during that day (accounting for grace period). Streak resets to 0 if the habit is missed beyond the grace window. Longest-ever streak is retained permanently regardless of resets.	**Must-Have**

GAM-802	Streak Display	Current streak shown prominently: on the user's group profile card, on the leaderboard row, and on the dashboard. A flame icon visually indicates active streaks. Streaks ≥ 7 days display with a gold flame. Streaks ≥ 30 days display with a diamond flame.	**Must-Have**
GAM-803	Badges & Achievements	Badges awarded automatically and displayed on user profile. Launch badge set: First Completion, 7-Day Streak, 30-Day Streak, 100-Day Streak, First Group Created, First Group Joined, Challenge Window Winner, Perfect Week (all habits every day for 7 days), Proof Pioneer (first proof photo in a new group).	**High**
GAM-804	Challenge Window Winner Declaration	At challenge window end: the member with the highest window score is declared the winner. A winner card is posted to the group activity feed. A 'Challenge Winner' badge is awarded. If there is a tie, tie-breaking logic (Section 8) applies.	**Must-Have**
GAM-805	Personal Stats Dashboard	Each user has a personal stats view (visible only to themselves): all-time completions per habit per group, current and longest streak per habit per group, completion rate over last 7 / 30 / 90 days, and a calendar heatmap of daily completion history.	**High**

5. System Architecture & Technical Design

5.1 Platform Targets

Platform	Technology	Notes
iOS	React Native (Expo managed workflow)	HealthKit-free. Camera via expo-camera. Token storage via expo-secure-store.
Android	React Native (Expo managed workflow)	Camera via expo-camera. Token storage via expo-secure-store.
Web	React Native Web (react-native-web)	Full feature parity. Camera via browser MediaDevices API (HTTPS required). HttpOnly cookies for session tokens.

5.2 High-Level Architecture

Rivals follows a standard client-server architecture. The React Native (Expo managed) app and web client communicate with a cloud-hosted backend over HTTPS REST API. Leaderboard and feed data refresh via polling (60-second interval). Proof photos are uploaded directly from the client to Cloudflare R2. Scheduled jobs run inside PostgreSQL via pg_cron. No Redis, no BullMQ, no separate Media Service — all infrastructure is chosen for zero cost at MVP scale.

Component	Technology	Responsibility
Mobile / Web Client	React Native + react-native-web	Cross-platform UI, camera capture, watermark rendering, state management, offline caching, push handling
API Server	Node.js + Fastify	Core business logic, auth, groups, habits, logs, leaderboard computation, polling endpoints
File Upload (R2)	Cloudflare R2 SDK (direct upload)	Proof photo upload direct to R2 (presigned URL from API). Server-timestamp validation in API. Zero egress fees.
Refresh Layer	React Query polling (60s)	Leaderboard and feed refresh every 60 seconds via React Query polling. No persistent WebSocket connection. Acceptable UX trade-off for zero infrastructure cost.
Database (Primary)	Supabase PostgreSQL (free tier)	All relational data: users, groups, memberships, habits, logs, streaks, badges, challenges. Auth via Supabase Auth. 500MB free tier.
Cache	React Query (client-side)	Client-side query cache (AsyncStorage on mobile, localStorage on web). No server-side Redis cache in MVP.
Background Jobs	Supabase pg_cron	Streak recalculation, challenge window evaluation, daily reminder dispatch, grace period expiry checks — all via SQL functions scheduled with pg_cron. No external queue.
Component	**Technology**	**Responsibility**
File Storage	Cloudflare R2	Proof photo storage, user avatars. Signed URLs for access control.
Push Notifications	Firebase Cloud Messaging + APNs	Cross-platform push delivery for reminders, streak alerts, leaderboard events

Auth	Custom JWT (or Firebase Auth)	Token issuance, refresh, Google OAuth callback, session management
Monitoring	Sentry + application metrics	Error tracking (client + server), API latency, sync health, job queue depths

5.3 Client Architecture

- **State Management:** Zustand for global client state. React Query (TanStack Query) for server state, caching, and background refetch.
- **Navigation:** React Navigation 6 — bottom tab navigator (Dashboard, Groups, Notifications, Profile) with nested stack navigators within each tab.
- **Offline Support:** React Query's persistent cache (AsyncStorage on mobile, localStorage on web) stores the last fetched feed, leaderboard, and habit status for read-only offline access. Write operations (habit logging) require network connectivity.
- **Real-Time:** React Query polling with a 60-second refetchInterval on leaderboard and feed queries. On app foreground, an immediate refetch fires. No persistent WebSocket connection in MVP — acceptable trade-off for zero infrastructure cost.
- **Camera (Mobile):** expo-camera (Expo managed workflow). Gallery access is not requested. Watermark rendered using expo-image-manipulator or HTML Canvas (via expo-gl) before upload.
- **Camera (Web):** MediaDevices.getUserMedia() with video constraint. Watermark burned onto a hidden HTML Canvas element. Resulting canvas exported as JPEG blob for upload.

5.4 Backend Architecture (Modular Monolith)

The backend is structured as a modular monolith — clear service boundaries within a single deployable, designed for easy decomposition if scale demands it.

- **Auth Module:** Registration, login, Google OAuth, JWT issuance, password reset, session revocation.
- **User Module:** Profile CRUD, avatar management, username uniqueness, account deletion.
- **Group Module:** Group lifecycle, membership management, invite link generation and validation, admin role transfer.
- **Habit Module:** Habit creation, editing, deactivation, grace period logic, completion state per user per day.
- **Log Module:** Habit log creation (proof photo upload via R2 presigned URL, server-timestamp validation), same-day edit/delete, log retrieval, daily completion status computation.
- **Leaderboard Module:** Score computation per mode, streak aggregation, challenge window logic. Results written directly to the leaderboard table; clients poll the API every 60 seconds.
- **Feed Module:** Feed event creation, pagination. Feed data served via polling endpoints; no push broadcast.

- **Notification Module:** Event-driven push dispatch, reminder scheduling, user preference management.
- **Gamification Module:** Badge evaluation and awarding, streak milestone detection.

5.5 Infrastructure & Deployment

- Containerized API deployed on Railway or Render. No Docker orchestration required for MVP — single service deployment.
- Separate staging and production environments with automated CI/CD pipeline.
- API server on Railway or Render (free tier for low traffic, $5/month for always-on). No separate Media Service — upload handled by API issuing R2 presigned URLs.
- Supabase PostgreSQL free tier (500MB). Automated backups included. No read replicas needed at MVP scale — connection pooler (PgBouncer, built into Supabase) handles concurrent polling load.
- Cloudflare R2 for proof photo storage (10GB free, zero egress fees). R2 presigned URLs with 1-hour TTL for access control. Cloudflare's global CDN serves the photos automatically.

[DEFERRED] Specific cloud provider for API hosting (Railway vs. Render vs. Fly.io) to be finalized at Phase 1 kickoff. Supabase and Cloudflare R2 are confirmed choices.

5.6 MVP vs. Scale Architecture

Rivals is built in two distinct infrastructure stages. The MVP stack targets zero infrastructure cost and ships the full product experience. The Scale stack is the target architecture once user numbers justify the additional cost. The data model and API contracts are identical in both stages — scaling is purely an infrastructure swap, not an architectural rewrite.

Layer	MVP Stack (Zero Cost)	Scale Stack (When Needed)
Database	Supabase free tier (500MB)	Supabase Pro ($25/mo) + read replica
Photo Storage	Cloudflare R2 (10GB free, zero egress)	Cloudflare R2 (scales, same pricing)
API Hosting	Railway / Render free tier	Railway Pro / Fly.io with auto-scaling
Refresh Model	React Query polling (60s interval)	WebSocket (Socket.io) for instant push
Cache	None (React Query client cache only)	Redis (leaderboard snapshots, rate limiting)
Background Jobs	Supabase pg_cron (SQL functions)	BullMQ on Redis (reliable async queues)

Migration Trigger	—	DB approaching 500MB, or polling latency complaints, or 50+ active groups

6. Data Model & Storage Strategy

6.1 Core Entities

Entity	Key Fields
users	id (UUID), username (UNIQUE, lowercase), email (UNIQUE), display_name, avatar_url, auth_provider (email | google), password_hash (nullable for Google), created_at, updated_at, timezone, deleted_at
groups	id, name, description, avatar_url, admin_user_id (FK → users), invite_code (UNIQUE), invite_link_enabled (bool), max_members (100), leaderboard_mode (streak | total | challenge), created_at
group_memberships	id, group_id (FK), user_id (FK), role (admin | member), joined_at, left_at (nullable), notification_prefs (JSONB)
habits	id, group_id (FK), name, description, grace_period_days (0 | 1 | 2), created_by (FK → users), created_at, is_active (bool), deactivated_at
habit_logs	id, habit_id (FK), user_id (FK), group_id (FK), log_date (DATE), completed_at (TIMESTAMPTZ), proof_photo_url, client_timestamp, server_timestamp, is_grace_day (bool), created_at, deleted_at (nullable — for same-day deletes)
streaks	id, user_id (FK), habit_id (FK), group_id (FK), current_streak, longest_streak, last_active_date (DATE), last_reset_at, updated_at
challenge_windows	id, group_id (FK), name, start_date (DATE), end_date (DATE), status (upcoming | active | completed), winner_user_id (nullable FK), created_by (FK), created_at
challenge_scores	id, challenge_window_id (FK), user_id (FK), completion_count, rank, computed_at
badges	id, user_id (FK), group_id (nullable FK), badge_type (ENUM), awarded_at, metadata (JSONB)

feed_events	id, group_id (FK), event_type (ENUM), actor_user_id (FK), target_user_id (nullable FK), habit_id (nullable FK), habit_log_id (nullable FK), metadata (JSONB), created_at
feed_reactions	id, feed_event_id (FK), user_id (FK), emoji, created_at — UNIQUE (feed_event_id, user_id)
notifications	id, user_id (FK), group_id (nullable FK), type (ENUM), title, body, data (JSONB), read_at, sent_at, created_at

6.2 Key Indexes

- habit_logs: (user_id, habit_id, log_date) — for daily completion checks
- habit_logs: (group_id, log_date) — for group-wide daily feed
- streaks: (user_id, habit_id, group_id) UNIQUE — for fast streak lookups
- feed_events: (group_id, created_at DESC) — for paginated feed queries
- group_memberships: (group_id, user_id) UNIQUE; (user_id) — for multi-group membershipqueries

6.3 Data Retention Policy

- Proof photos retained for the lifetime of the group. On group deletion, all photos are queued fordeletion within 90 days.
- On user account deletion: photos deleted, logs anonymized (user_id replaced with a tombstone),profile data purged within 90 days.
- Habit logs are never hard-deleted except as part of same-day re-submission (soft delete viadeleted_at) or account deletion.
- OAuth tokens stored encrypted at rest, never logged, proactively refreshed before expiry.

6.4 Caching Strategy

- **No server-side Redis cache in MVP.** Leaderboard scores computed and stored in PostgreSQL on each log event. Supabase PgBouncer handles concurrent polling load.
- **React Query (client):** Feed, leaderboard, and habit status cached with configurable stale times. Background refetch on app foreground and on 60-second polling interval.
- **CDN:** Proof photos and avatars served via Cloudflare R2 with presigned, time-limited URLs (1-hour TTL). Zero egress fees.

7. Proof Photo & Anti-Cheat System — Deep Dive

This section details the full technical and UX flow of the proof photo system, which is the single most critical differentiator of Rivals. Every design decision here is made to maximize integrity while minimizing friction for honest users.

7.1 Completion Flow (Step by Step)

Step 1: User taps 'Complete' on a habit card.

The app checks whether the habit has already been completed today. If yes, shows an option to re-submit (same-day edit). If no, proceeds.

Step 2: App opens the device camera.

Camera opens in photo mode only. No gallery button is shown. On mobile, the app requests camera permission (not photo library). On web, getUserMedia() is called with HTTPS.

Step 3: User takes the photo.

User frames and captures. A confirmation screen shows the preview.

Step 4: Watermark is burned onto the image.

Client-side: a canvas rendering layer overlays the watermark text (date, time, username, habit name) directly onto the image pixels. The resulting image with the watermark is what gets uploaded — the clean original is discarded.

Step 5: Client sends upload request to the API.

Request includes: watermarked JPEG, client timestamp (ISO 8601 with timezone), habit_id, group_id, user_id (from JWT). API generates an R2 presigned URL, client uploads directly to R2, then notifies the API with the storage URL.

Step 6: API records server timestamp.

At the moment the request is received, the server records its own UTC timestamp.

Step 7: Timestamp delta check.

If |server_timestamp - client_timestamp| > 5 minutes, upload is rejected. Error message: 'Photo timestamp does not match. Please ensure your device clock is correct.'

Step 8: Photo stored and log created.

Photo is stored in Cloudflare R2. A habit_log record is created in Supabase with both timestamps. The API synchronously recalculates the user’s leaderboard score and inserts a feed event record.

Step 9: Checkbox is auto-checked.

The habit card on the user's dashboard updates to 'Completed' with a green checkmark and the proof photo thumbnail.

Step 10: Real-time push to group.

No WebSocket push in MVP. Other group members will see the updated leaderboard and the new feed card on their next 60-second polling cycle, or immediately if they pull-to-refresh.

7.2 Watermark Specification

Property	Value
Position	Bottom-left corner, 16px padding from edges
Content	Line 1: DD/MM/YYYY HH:MM:SS (local time) | Line 2: @username | Line 3: Habit Name
Font	Monospace, 14px, semi-bold
Color	White text with 1px dark stroke (for visibility on any background)
Background	Semi-transparent dark pill behind each line (opacity 0.55)
Rendering	Burned into image pixels before upload. Canvas toDataURL() / toBlob() used client-side.
Removal Resistance	Watermark is part of the JPEG pixel data after rendering. It cannot be removed without visible image degradation.

7.3 Anti-Cheat Limitations & Accepted Trade-offs

The proof system significantly raises the cost of cheating but is not cryptographically unbreakable. The following limitations are accepted in v1.0:

- **Device clock manipulation:** The 5-minute server delta check mitigates most clock attacks. A user who sets their clock back ≥5 minutes will have their upload rejected.
- **Photo of a photo:** A user could take a photo of a screen displaying a gym image. This is a social trust violation detectable by group members reviewing the proof feed — the group polices itself.
- **VPN / proxy timestamp spoofing:** Server-side timestamp is always used as the authoritative reference. Client-reported time is for the watermark display only; the server never trusts the client timestamp for validation purposes.
- **Screen recording / screenshot workarounds:** The same 'photo of a screen' limitation applies.

Accepted in v1.0; future versions may add GPS or device sensor checks.

[DEFERRED] *GPS metadata or device sensor verification (accelerometer proof of movement) may be added in a future version as optional per-habit settings.*

8. Leaderboard Engine & Scoring Logic

This section precisely defines how each leaderboard mode computes scores and rankings. The admin can switch between modes at any time. Switching never deletes data from other modes — all historical data is retained and the inactive modes can be re-activated with full history.

8.1 Mode 1: Streak Mode

What is displayed: Members ranked by current active streak (consecutive days with all habits completed).

Streak Rules:

- A streak day requires ALL active habits in the group to be completed with a valid proof photo.
- The streak increments once per calendar day (in the user's local timezone). Completing habitsmultiple times in one day does not increment more than once.
- If a habit has a grace period of N days: the streak is protected for N days after a missed day. Ifthe user logs within the grace window, the streak is maintained. If not, the streak resets to 0 on the day the grace period expires.
- Longest-ever streak is retained permanently. It is displayed on the profile and used as atie-breaker.

Score: current_streak (integer, days)

Tie-breaking: (1) longest_ever_streak DESC, (2) total_all_time_completions DESC, (3) joined_at ASC (earlier member wins)

8.2 Mode 2: Total Completions Mode

What is displayed: Members ranked by total number of habit completions since they joined the group (all habits, all time).

Scoring Rules:

- Each accepted proof photo log = 1 completion point.
- A group with 2 habits awards up to 2 points per day per member.
- Grace period completions count as valid completions.
- If a member leaves and rejoins, their historical completions (from before they left) are excludedfrom the running total for fairness. Completions from the date of rejoin onwards count.

Score: total_completions (integer, count of accepted proof logs)

Tie-breaking: (1) current_streak DESC, (2) longest_ever_streak DESC, (3) joined_at ASC

8.3 Mode 3: Challenge Window Mode

What is displayed: Members ranked by completions within the current (or most recent) admin-defined time window.

Window Rules:

- Admin creates a challenge window with: name (e.g., 'October Grind'), start date, end date. Minimum window: 3 days. Maximum window: 365 days.
- Only completions timestamped between start_date 00:00:00 and end_date 23:59:59 (in thegroup's reference timezone, set at group creation) count toward the window score.
- At window end: (1) final scores are frozen, (2) winner is declared, (3) a Winner badge is awarded, (4) a winner announcement is posted to the group feed, (5) the window is archived.
- Admin can create a new window after the previous one ends. Windows cannot overlap. A newwindow cannot start while one is active.
- Historical windows are archived and viewable by group members via a 'Past Challenges' tab.
- Max completions per day per habit = 1 (grace period completions count if within the windowdates).

Score: window_completions (integer, completions within the active window period)

Tie-breaking: (1) current_streak DESC, (2) longest_ever_streak DESC, (3) joined_at ASC

8.4 Mode Switching Behavior

Action	Effect
Admin switches from Streak to Total Completions	Leaderboard immediately re-renders using total completions data. Streak data is preserved in the database and will be restored if admin switches back.
Admin switches from any mode to Challenge Window	If no active window exists, a prompt asks admin to configure one. If a window exists, the leaderboard shows the current window score.
Admin switches away from Challenge Window mid-window	The active window is paused (completions stop being counted toward it). If admin returns to Challenge Window mode, the window resumes. Admin can also end the window early.
Admin ends a challenge window early	Final scores frozen as of the moment admin triggers 'End Window'. Winner declared immediately. Same winner flow as natural end.

8.5 Leaderboard Computation Architecture

- On each accepted habit log: the Leaderboard Module synchronously recomputes the affected user's score for all modes (streak, total, window) and writes the updated scores to the leaderboard_scores table in PostgreSQL.
- The leaderboard_scores table is the source of truth. The GET /groups/:id/leaderboard endpoint queries it directly with a simple ORDER BY score DESC. At MVP scale (100 members per group, polling every 60s) this is fast enough without a cache layer.
- Scheduled jobs (pg_cron): midnight grace period expiry check runs daily. Challenge window end evaluation runs every hour. Daily reminder dispatch runs at the user's configured reminder time via a cron function.
- Clients receive updated leaderboard data on their next 60-second poll. Pull-to-refresh on the leaderboard screen triggers an immediate refetch. No WebSocket broadcast in MVP.
- Clients receiving the event trigger a React Query cache invalidation, causing a silent backgroundrefetch and re-render.

9. User Experience & Design

9.1 Design Principles

- **Proof-first UX:** The camera should feel like a natural, fast part of the completion flow, not a hurdle. Opening to the camera should take one tap.
- **Live feedback:** Leaderboard changes, new proof photos, and streak milestones should feel immediate. Animations reinforce the competitive energy.
- **Group-centric navigation:** A user's primary context is always their group. The dashboard surfaces group status prominently.
- **Dark mode as default:** Warm amber accent on deep navy background. Light mode available.
- **Zero ambiguity:** It must always be obvious which habits are done, which are pending, and where you stand on the leaderboard.

9.2 Navigation Structure

Tab	Content
Dashboard	Today's habit completion status across all groups. Quick-complete buttons per habit. Streak counters. Group rank previews.
Groups	List of all joined groups with rank preview and today's completion status. Tap to enter a group (Leaderboard | Feed | Habits | Members).
Notifications	All notification types in reverse chronological order. Mark all as read. Tap to navigate to the relevant group or event.
Profile	Display name, avatar, username, badges. Personal stats per group. Account settings. Notification preferences.

9.3 Core User Journeys

Journey 1: New User Invited via Link

Taps invite link → App Store / web if not installed → Sign-up (email or Google, choose username) → Lands on group join confirmation screen → Accepts → Sees group dashboard with today's habits and the leaderboard → Taps 'Complete' on first habit → Camera opens → Takes photo → Watermark appears on preview → Confirms → Habit marked complete → Leaderboard updates live → Feed shows their proof photo to all group members.

Journey 2: Daily Returning User

Opens app → Dashboard shows which habits are done / pending across all groups → For pending habits: taps 'Complete', camera opens, photo taken, watermark confirmed, done → Checks leaderboard to see if anyone moved → Scrolls feed to see friends' proof photos → Closes app.

Journey 3: Admin Creating a Challenge Window

Navigates to group → Group Settings → Leaderboard Mode → Selects 'Challenge Window' → Taps 'New Challenge' → Names it 'November Grind', sets start Nov 1, end Nov 30 → Confirms → Group leaderboard switches to window mode with a countdown timer → All members receive a push notification about the new challenge.

9.4 Edge Cases & UI Rules

- **No habits in group yet:** Group dashboard shows an 'Admin needs to add habits' empty state. Non-admin members see this with a note that the admin will set habits.
- **Habit completed today:** Completion card shows proof photo thumbnail, timestamp, green checkmark. 'Re-submit' button available until midnight.
- **Camera permission denied:** Full-screen prompt explaining that camera access is required for proof and how to enable it in device settings.
- **Upload fails (network error):** Photo is held in a local queue and retried automatically when connectivity is restored. User sees a 'Saving...' indicator.
- **Admin tries to leave without transferring:** 'Leave Group' button is disabled with a tooltip: 'Transfer admin role before leaving.'
- **Grace period expiry today:** A banner on the habit card shows 'Grace period ends tonight at midnight. Log now to protect your streak.'
- **Challenge window ends in 24 hours:** A countdown banner appears on the leaderboard screen for all members.

[DEFERRED] Full design system (color tokens, typography scale, component library, spacing grid) to be built in Phase 1 by the design lead.

10. Privacy, Security & Compliance

Rivals handles proof photos, personal health-adjacent behavior data, and location-agnostic timestamps. Privacy and security are non-negotiable requirements, not post-launch additions.

ID	Requirement	Description	Priority

SEC-1001	TLS 1.2+ Everywhere	All API calls, WebSocket connections, S3 presigned URL requests, and push notification payloads must use TLS 1.2 or higher. No exceptions, including internal service-to-service calls.	**Critical**
SEC-1002	AES-256 Encryption at Rest	All sensitive database fields (OAuth tokens, refresh tokens, email addresses) encrypted at rest using AES-256. PostgreSQL column-level encryption or transparent disk encryption at minimum.	**Critical**
SEC-1003	OAuth Token Security	Third-party OAuth tokens (Google) stored encrypted, never logged, proactively refreshed 10 minutes before expiry, scoped to minimum permissions (email + profile only).	**Critical**
SEC-1004	Proof Photo Access Control	Proof photos are stored in private S3 buckets. Client access is via signed URLs with a 1-hour TTL. Photo URLs are never stored in the client cache beyond the TTL. Photos are accessible only to members of the group to which the log belongs.	**Critical**
SEC-1005	Username Privacy	Usernames are the only identifier visible to other users. Email addresses are never exposed through any API endpoint to any other user.	**Must-Have**
SEC-1006	Proof Photo Visibility Scope	Proof photos are visible only within the group. They cannot be downloaded, shared via the app, or accessed without group membership. The app does not provide a download button for proof photos.	**Must-Have**
SEC-1007	Data Export (User Right)	Users can request a full export of their data: profile info, all habit logs (with metadata but not photo blobs), streak history, and badge history. Delivered within 7 days as a JSON archive.	**Must-Have**
SEC-1008	Account Deletion	Full account deletion on request. Personal data purged within 30 days. Habit logs anonymized (user_id tombstoned). Proof photos deleted. Group membership records removed.	**Must-Have**

ID	Requirement	Description	Priority

SEC-1009	GDPR & CCPA Readiness	Right to access, right to deletion, right to portability, explicit consent at registration. Privacy policy and Terms of Service hosted on company-owned domain, linked in-app and in store listings.	**Must-Have**
SEC-1010	Rate Limiting	API rate limiting on all endpoints, especially auth (login attempts, password reset) and habit log submission (max 10 attempts per habit per day per user). In-memory rate limiter for MVP; migrate to Redis-backed limiter when scaling to multiple API instances.	**Must-Have**
SEC-1011	Input Validation & Injection Prevention	All user-generated text fields sanitized and validated server-side. Parameterized queries throughout (Drizzle ORM / pg). No raw SQL string concatenation.	**Critical**

11. Monetization Strategy

11.1 Pricing Model: Free Forever

Rivals is completely free for all users for now. There are no subscription tiers, no premium features, no in-app purchases, and no paid upgrades. Every user and every group gets full access to every feature: unlimited groups, unlimited habits, all leaderboard modes, challenge windows, the full proof photo system, and all gamification features.

This is a deliberate strategic choice. Rivals' value is entirely dependent on network density: a group where half the members can't afford a paid tier is a broken group. Keeping the app 100% free maximizes the social loop and ensures every invited friend can participate fully.

> ***[DEFERRED]*** *Revenue model to be revisited after 10,000 active users. No monetization without transparent community communication first.*

13. Development Phases & Milestones

Seven phases take Rivals from a blank project to a fully testable, deployable product. Deployment to the App Store, Google Play, and web hosting comes after Phase 7. Each phase has a clear exit condition you can verify before moving on.

Tech Stack: React Native (Expo managed workflow), Node.js + Fastify, Supabase (PostgreSQL + Auth), Cloudflare R2, React Query (polling), Zustand, Drizzle ORM, Firebase Cloud Messaging + APNs, Supabase pg_cron, Sentry — organized in a pnpm monorepo.

Phase 1 — Foundation & Infrastructure

Goal: Stand up the complete project skeleton so every future phase has a stable, runnable base.

Deliverables

- pnpm monorepo scaffolded: apps/mobile (React Native Expo managed), apps/web (React Native Web via Expo for Web), packages/api (Fastify), packages/shared.
- PostgreSQL provisioned and connected. Full database schema migrated for ALL core entities(Section 6.1).
- Supabase project provisioned: PostgreSQL connected, Auth configured (email + Google), Storage bucket created for avatars.

- Auth system end-to-end: email/password sign-up, Google Sign-In, username uniquenessenforcement, email verification, JWT issuance, refresh token rotation, logout from all devices.
- React Navigation: bottom tabs (Dashboard, Groups, Notifications, Profile) with placeholderscreens on all platforms.
- Zustand global state and React Query initialized. Offline cache foundation.
- User profile screen: display name, avatar upload, username display, account management.
- Staging and production environments separated. CI/CD pipeline operational.
- TLS enforced end-to-end from day one.

Exit Condition

App launches on iOS simulator, Android device, and web browser. User can sign up (email + Google), log in, navigate all four tabs, view their profile, and log out from all devices. Database schema fully migrated.

Phase 2 — Groups & Habit Management

Goal: Build the group layer — the fundamental social unit of the product.

Deliverables

- Create Group flow: name, description, avatar.
- Invite by username: admin searches username, sends invite, target user receives in-appnotification and can accept/decline.
- Invite link + alphanumeric code: generation, sharing, join flow. Deferred deep linking: UniversalLinks (iOS) and App Links (Android) configured.
- Group dashboard: member list, group settings, admin controls (remove member, edit group,transfer admin role).
- Admin leave guard: Leave button disabled until admin role is transferred.
- Multi-group membership: user can be in multiple groups. Groups tab shows all groups with basicstats.
- Habit creation by admin: name, description, grace period (0/1/2 days), activate/deactivate.
- Habit display on group dashboard: today's status (complete / pending / grace period active).

Exit Condition

Admin creates a group, invites a user by username, the user accepts. Admin creates two habits. Both users see the habits on the group dashboard. Admin cannot leave without transferring role.

Phase 3 — Proof Photo & Habit Logging

Goal: The proof system — the core differentiator. This phase must be bulletproof.

Deliverables

- Live camera integration: camera-only mode, gallery blocked, on all three platforms.
- Timestamp watermark rendering: client-side canvas burn-in with date, time, username, habitname.
- API generates R2 presigned URLs for direct client upload. Server timestamp recorded on upload notification. Validates delta (≤5 min), stores to S3, creates habit_log record.
- Same-day re-submission: delete and re-submit within same calendar day (local timezone).
- Log locking at midnight: no edits or backdating after day ends.
- Habit card UI: pending / completed / grace states with proof thumbnail on completion.
- Grace period logic: streak protection during grace window, reset on expiry.
- Web camera proof: browser MediaDevices API, canvas watermark, HTTPS-only.

Exit Condition

User taps Complete, camera opens (no gallery button), photo is taken, watermark is visible on preview, submission goes through, habit marked complete. Uploading a photo more than 5 minutes from server time is rejected. After midnight, logs from today are locked.

Phase 4 — Leaderboard & Real-Time Engine

Goal: Bring the competitive engine live with all three modes and real-time updates.

Deliverables

- Leaderboard Module: score computation for Streak, Total Completions, and Challenge Windowmodes.
- Leaderboard Module: synchronous score recomputation on each log event, written to leaderboard_scores table in PostgreSQL. pg_cron job for challenge window end evaluation.
- Polling layer: React Query refetchInterval set to 60 seconds on leaderboard and feed queries. Pull-to-refresh wired on all group screens.
- Leaderboard UI: ranked list with avatar, display name, score, delta, current user highlighted. Re-renders on each 60-second poll cycle or pull-to-refresh.
- Tie-breaking logic implemented and displayed in tooltip.
- Challenge Window management: admin creates window (name, start, end), countdown timer,score tracking, winner declaration, badge award, historical archive.
- Admin mode-switching: Group Settings → Leaderboard Mode selector.

- Minimum Members Gate: 'Waiting for Rivals' state with invite CTA.

Exit Condition

Two users in a group. Both submit proof photos. Leaderboard updates live within seconds. Switch modes — leaderboard re-renders correctly for each mode. Admin creates a challenge window, both users log, window ends, winner is declared.

Phase 5 — Feed, Notifications & Gamification

Goal: Complete the social and engagement loop.

Deliverables

- Group Activity Feed: real-time event cards (proof photo submissions, streak milestones, memberjoins, challenge events), inline photo thumbnails, full-size photo viewer.
- Emoji reactions on feed cards (single reaction per user per card).
- Push notification infrastructure: FCM + APNs. Notification Module wired to all event types(Section 4, EPIC 7).
- Daily habit reminder at user-configured time. Streak-at-risk notification 2 hours before gracedeadline.
- Notification preferences screen: per-type, per-group toggles.
- Streak tracking: current and longest streak per habit per group. Streak display on leaderboardrows, profile, and dashboard.
- Badge system: all launch badges evaluated and awarded on relevant events.
- Personal stats view: completion rate, streak history, calendar heatmap per habit per group.
- Gamification Module: streak milestone detection and milestone feed events.

Exit Condition

User submits proof photo → appears in group feed within 3 seconds → other members receive push notification (if enabled). User hits 7-day streak → badge awarded, milestone card appears in feed. Reminder fires at configured time for users with incomplete habits.

Phase 6 — Web Parity & Polish

Goal: Ensure the web platform has full feature parity with mobile and refine the UX across all platforms.

Deliverables

- React Native Web: audit every screen for web rendering correctness (responsive layout, nomobile-only components leaking).
- Browser camera proof flow: getUserMedia, canvas watermark, upload — full functional parity withmobile.
- Web push notifications via Web Push API (service worker) or FCM Web SDK.
- Deep link onboarding for web: invite link lands on join screen in browser, handlesunauthenticated users gracefully.
- Responsive layout: web dashboard usable on desktop (1280px+), tablet (768px), and mobile web(375px).
- Cross-platform UX polish: loading states, error states, empty states, skeleton screens on all keyviews.
- Onboarding flow: step-by-step walkthrough for new users (sign up → join/create group → first habit log). Skip and return later option.
- Contextual tooltips for first-time users on all key UI elements.

Exit Condition

A brand new user clicks an invite link in a desktop browser, signs up, joins the group, completes a habit proof via browser camera, and sees the leaderboard update — all without installing any app.

Phase 7 — Security, Hardening & Pre-Deployment

Goal: Harden the full stack, close every edge case, instrument all metrics, and validate end-to-end.

Deliverables

- Security audit: TLS enforcement verified, encryption at rest confirmed, OAuth token storageaudited, rate limiting stress-tested.
- Proof photo access control audit: signed URL TTL verified, no unauthorized access paths.
- GDPR/CCPA flows: data export (JSON archive, ≤7 days delivery), account deletion cascade, consent at registration.
- Privacy policy and Terms of Service finalized and published on company-owned domain.
- Performance: database query optimization (all indexes per Section 6.2), leaderboard query EXPLAIN analysis, load test (500 concurrent users submitting proof photos simultaneously).
- Error tracking: Sentry configured for both mobile client and API server.
- Analytics instrumentation: all KPIs from Section 12 tracked. Analytics platform integrated(Mixpanel, Amplitude, or PostHog).
- End-to-end user journey testing: all three personas simulated against staging environment.
- Edge case testing: grace period expiry, challenge window end, admin transfer, multi-groupisolation, same-day re-submission lock.

- App Store and Google Play metadata prepared: screenshots, descriptions, HealthKit/Camerausage justifications.

Exit Condition

Full end-to-end test on iOS physical device, Android physical device, and Chrome desktop browser against staging. All three user journeys complete successfully. Security checklist signed off. Analytics dashboard live. No P0 or P1 bugs open. App is ready for deployment.

Phase Summary Table

Phase	Name	Primary Epics / Sections
1	Foundation & Infrastructure	AUTH-101–106, GRP-201–208 (scaffold), Sec 5.5
2	Groups & Habit Management	GRP-201–208, HAB-301–306
3	Proof Photo & Habit Logging	PHT-401–407, HAB-304–305, Sec 7
4	Leaderboard & Real-Time Engine	LB-501–507, Sec 8, Sec 5.3–5.4
5	Feed, Notifications & Gamification	FEED-601–605, NOTIF-701–708, GAM-801–805
6	Web Parity & Polish	Sec 9, ONB (full), Sec 5.1 Web
7	Security, Hardening & Pre-Deployment	Sec 10, Sec 12, Sec 15, edge cases

14. Future Roadmap (Post v1.0)

Feature	Description	Tentative Phase
Non-Daily Habit Frequency	Admin can set habits to 3x/week, weekdays only, custom schedule.	Post-launch v1.1
Member Habit Suggestions	Members can suggest new habits to the admin for approval.	Post-launch v1.1
GPS Verification (Optional)	Per-habit GPS check: user must be within a configurable radius of a set location (e.g., a gym address) to complete the habit.	v1.2+
Username Change	Allow users to change username once every 90 days.	v1.1
Public Groups / Discovery	Opt-in public group directory browsable by category.	v1.2+
Partial Credit (Multi-Habit Streaks)	Admin can configure streak to require only N of M habits (e.g., 2 of 3) rather than all.	v1.2+
Text Comments on Feed Cards	Lightweight commenting on proof photo feed cards.	v1.2+
Group Chat	Lightweight in-group messaging. Reactions and short messages only.	v2.0
In-App Proof Viewing Gallery	Per-member photo gallery of all their historical proof photos in a group.	v1.2+
Streak Freeze (Admin Option)	Admin can grant group members a limited number of streak freezes.	v1.2+
Habit Templates	Pre-built habit packs (Morning Routine Pack, Fitness Pack, etc.) for fast group setup.	v1.2+
iOS & Android Widgets	Home screen widget showing today's habit completion status and leaderboard rank.	v2.0

Wearable Integration	Optional: link Apple Watch or Fitbit step count as supporting evidence alongside the proof photo.	v2.0+
AI Habit Insights	Weekly AI-generated summary: consistency patterns, encouragement, group trend analysis.	v2.0+
Data Export (Full)	Full data export including proof photo archive as a ZIP.	v1.2+

15.2 Key Assumptions

- Expo managed workflow (expo-camera) supports blocking gallery access and returning raw image data for client-side watermark processing without ejecting to bare workflow.
- Browser MediaDevices API is sufficient for web camera proof on all major browsers (Chrome,Safari, Firefox) under HTTPS.
- Client-side canvas watermark rendering is performant enough on low-end Android devices to notnoticeably delay the proof submission flow (<1 second for watermark + compression).
- The group invite loop is the sole user acquisition channel. No paid marketing budget.
- 100 members per group is sufficient for the target use case (friend groups, small communities)and manageable for real-time leaderboard performance without sharding.

15.3 Risks & Mitigations

Risk	Description	Severity	Mitigation
Camera API fragmentation	Android device diversity may cause camera behavior inconsistencies (aspect ratios, front/back camera switching, gallery blocking effectiveness).	High	Extensive device testing matrix. Use battle-tested library Extensive device testing matrix. Use expo-camera with Expo managed workflow. Eject to bare workflow only if a native camera capability is unavailable in Expo managed.
Watermark bypass	Motivated cheaters could photograph a screen showing a desired image.	Medium	Accepted in v1.0. Group social policing. Future: GPS or sensor verification.
Server clock delta attacks	User sets device clock forward, takes photo, resets clock before upload.	Medium	5-minute server-side delta check. Clock manipulation beyond 5 minutes will fail the upload.

Risk	Description	Severity	Mitigation
Real-time scalability	Polling load on Supabase free tier. 100 members per group all polling every 60 seconds across many groups could stress the free tier (500MB DB, connection limits).	Medium	Supabase PgBouncer connection pooler handles concurrent polling. Keep leaderboard queries fast with proper indexes (Section 6.2). Upgrade to Supabase Pro ($25/month) when DB approaches 500MB or connection limits.
Cold start / empty groups	New groups with <2 members see no leaderboard. Early users may churn.	High	Minimum Members Gate (LB-506). Sample data in onboarding. Public group discovery in v1.2.
Web camera limitations	Some browsers (notably Safari on iOS) have limited or inconsistent camera API support.	Medium	Graceful degradation with clear error messaging. iOS users strongly guided to the native app for proof submission.

17. Appendix

A. Narrative: A Morning with Rivals

It is 6:47 AM on a Wednesday. Aryan's phone buzzes.

"Rohan just submitted his gym proof. He's now 3 days ahead of you."

Aryan was already awake. Now he is out of bed.

He pulls on his shoes, drives to the gym, and gets there by 7:15. He taps the Rivals app, opens the Morning Gym habit, and hits Complete. The camera opens. He points it at the weight rack, taps the shutter. The watermark burns in automatically: 07:17:34, 04/11/2026, @aryan_k, Morning Gym. He confirms the photo. The habit card flips to green.

In the group, 'The Brotherhood', the activity feed pings. Aryan's proof photo appears. Rohan sees it. He sends a fire emoji reaction.

By 8 AM, all six members of The Brotherhood have logged their gym proof. The leaderboard refreshes. Aryan is back in second. Shreya, who has not missed a single day in 34 days straight, sits unmoved at first.

Nobody is going to catch her. But everyone is going to try.

This is Rivals. The habit doesn't care if you're motivated. But your friends will notice if you're not.

B. Glossary

Term	Definition
Proof Photo	A live camera photo taken at the moment of habit completion, with a timestamp watermark burned into the image, used as evidence of completion.
Watermark	Text overlay burned into proof photo pixels showing date, time, username, and habit name. Cannot be removed without image degradation.
Streak	A count of consecutive days on which all active habits in a group were completed with valid proof photos.

Grace Period	An optional 1–2 day buffer after a missed day during which a user can still log and protect their streak.

Term	Definition
Challenge Window	An admin-defined time-boxed competition within a group (start date + end date). Members ranked by completions during the window. Winner declared at end.
Leaderboard Mode	One of three ranking systems: Streak Mode, Total Completions Mode, or Challenge Window Mode. Admin selects and can change at any time.
Group Isolation	Each group is fully independent. Habits, logs, streaks, and leaderboards from one group have no effect on any other group, even for the same user.
Admin	The group creator or a member to whom admin role has been transferred. Has sole authority over habit creation, member management, and leaderboard configuration.
Server Delta Check	A server-side validation that rejects proof photo uploads where the client-reported timestamp differs from the server-received timestamp by more than 5 minutes.
D7 / D30 Retention	The percentage of users who return to the app 7 or 30 days after first sign-up.

End of Document — Master PRD v1.0 — Rivals